‘Family Risk Sharing’
‘When the Shock Hits the Knot: Individual Consumption Insurance Among Spouses’
‘When the Shock Hits the Knot: bargaining and family risk sharing’

B-C-V

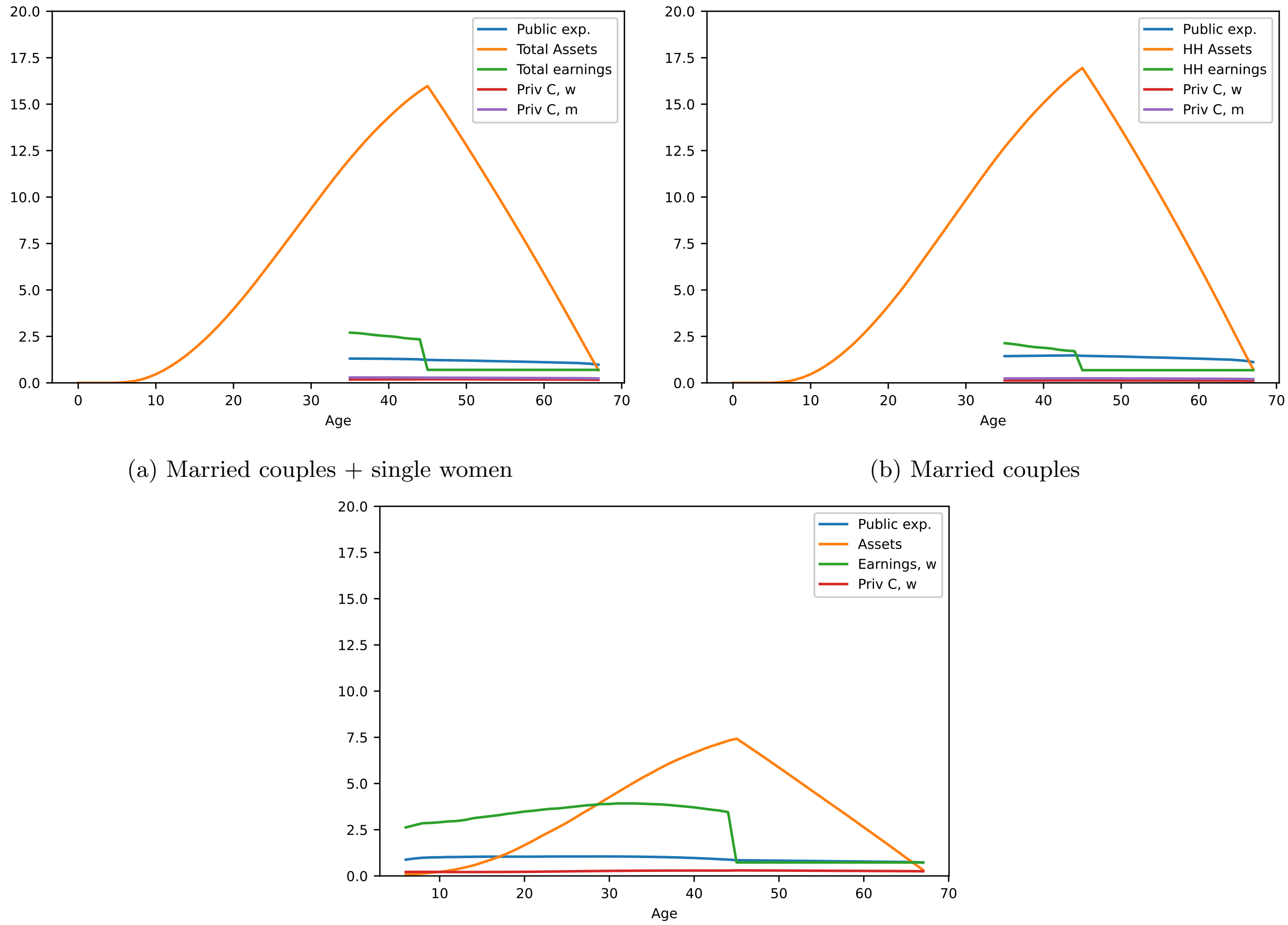
July 10, 2025

1 Summary statistics and life-cycle behavior

Table 1: Summary statistics

	Household assets (1)	Household earnings (2)	Wife, Private consumption (3)	Husband, Private consumption (4)	Home good expenditure (5)
Mean	5.977	nan	nan	nan	nan
Gini	0.572	nan	nan	nan	nan
Top 1% share	0.050	nan	nan	nan	nan

NOTES: assets and earnings are measure across the population regardless of marital status, while other variables are measured among married households.



(a) Married couples + single women

(b) Married couples

(c) Single women

Figure 1: Life-cycle behavior of different types of household, averages

2 Log consumption and income growth

Table 2: Moments of the log growth of the variables reported in the rows

	Mean	Variance	Skeweness	Kurtosis
Wife, private consumption	nan	nan	nan	nan
Husband, private consumption	nan	nan	nan	nan
Wife share of private consumption	nan	nan	nan	nan
Home good expenditure	nan	nan	nan	nan
Total consumption	nan	nan	nan	nan
Wife, earnings	nan	nan	nan	nan
Husband, earnings	nan	nan	nan	nan

NOTES: sample of those who stay married over two consecutive periods.

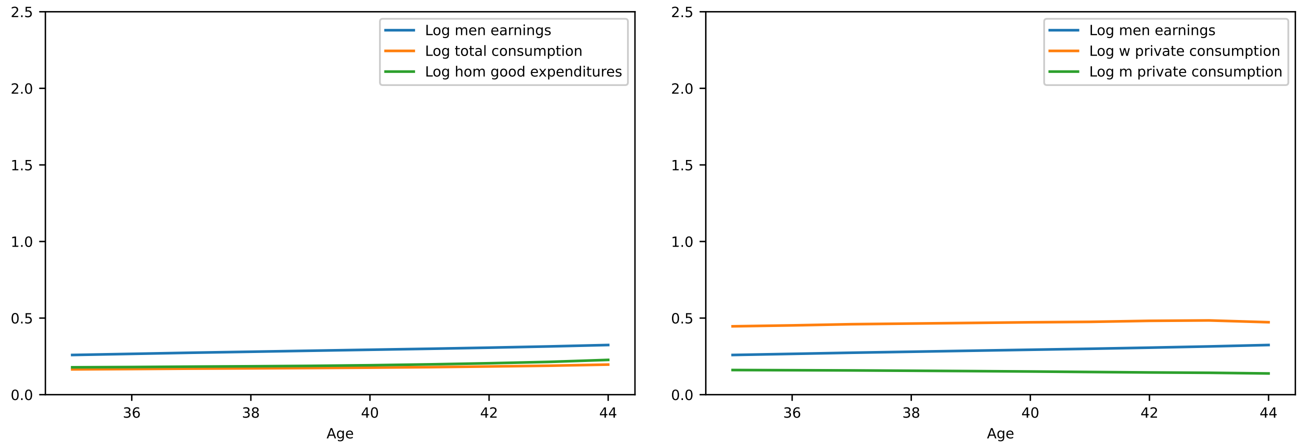
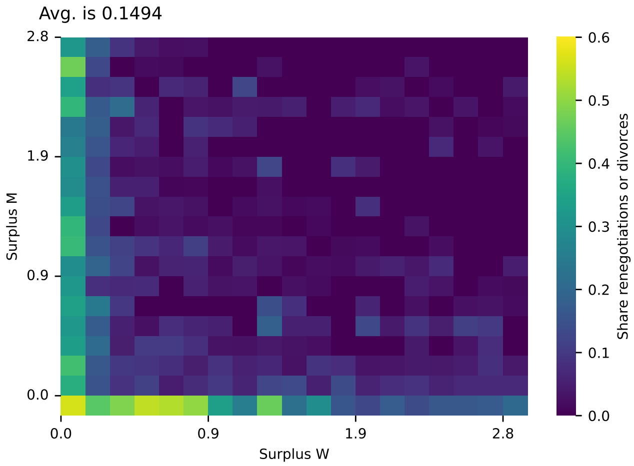
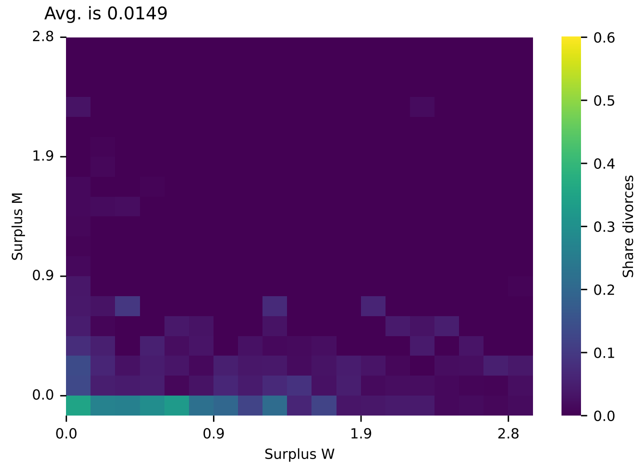


Figure 2: Variance of log earnings and consumption by age

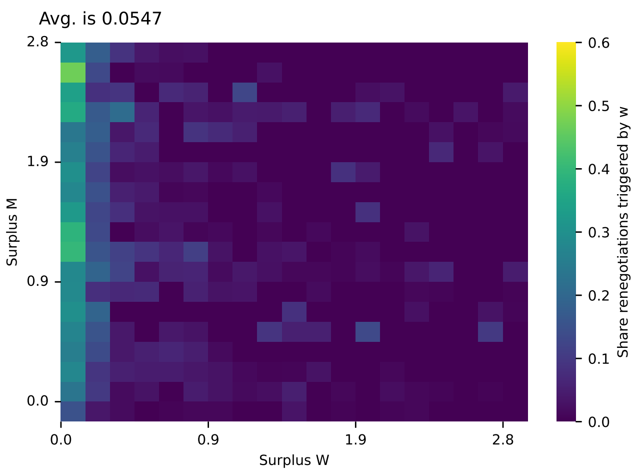
3 Marital surplus, renegotiation and divorce



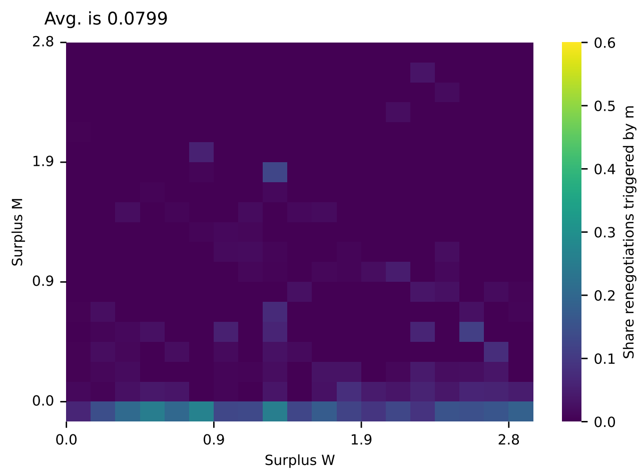
(a) Renegotiations and divorces



(b) Divorces



(c) Renegotiations triggered by the wife



(d) Renegotiation triggered by the husband

Figure 3: Marital surplus, renegotiation and divorce

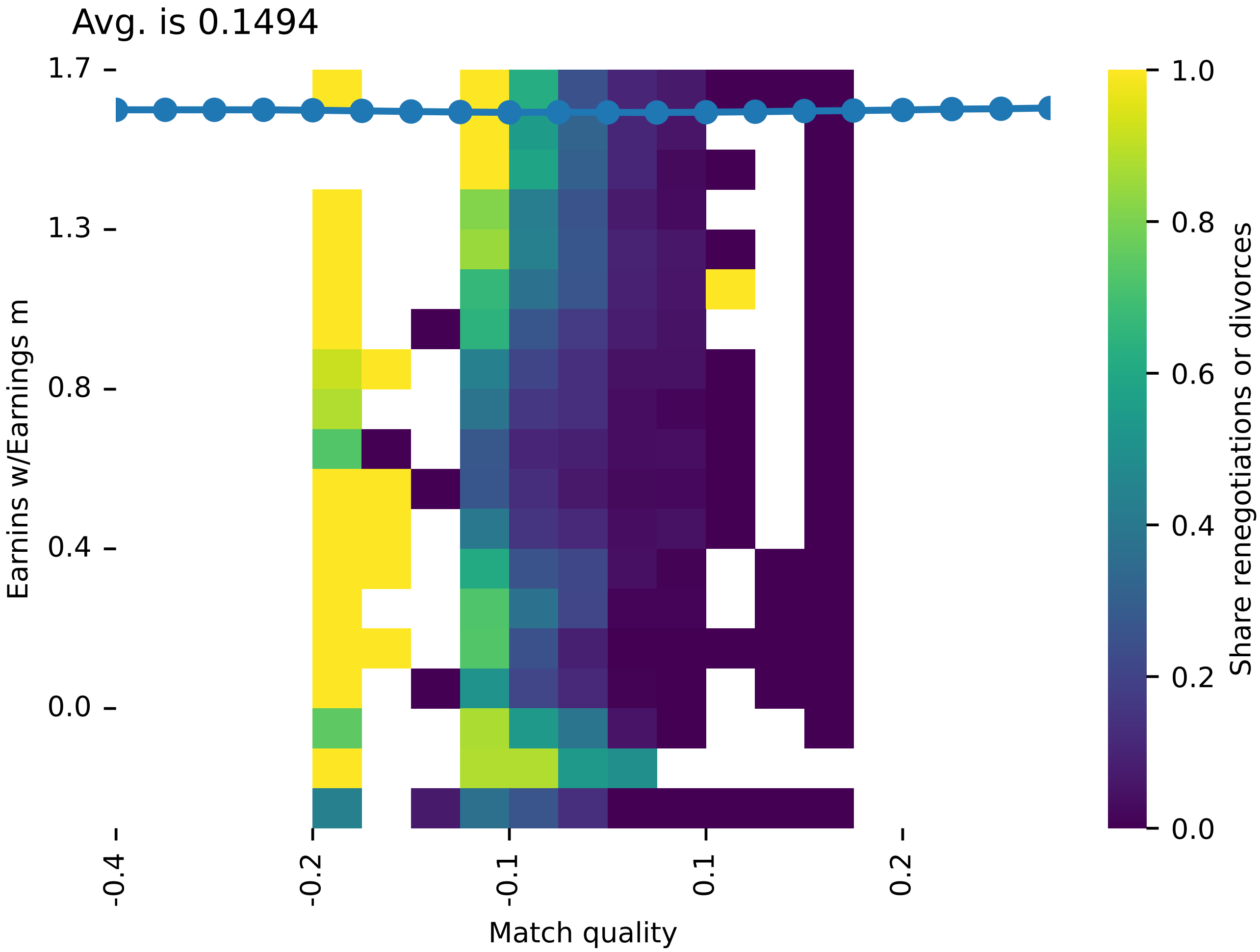
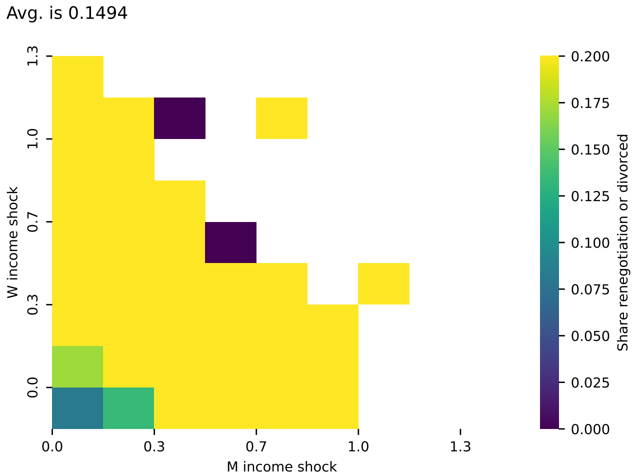
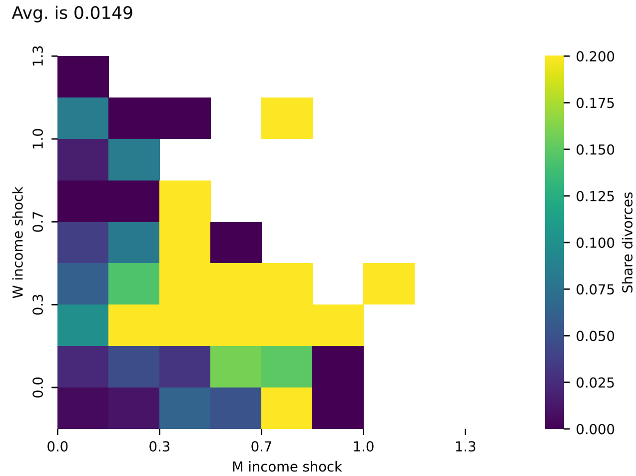


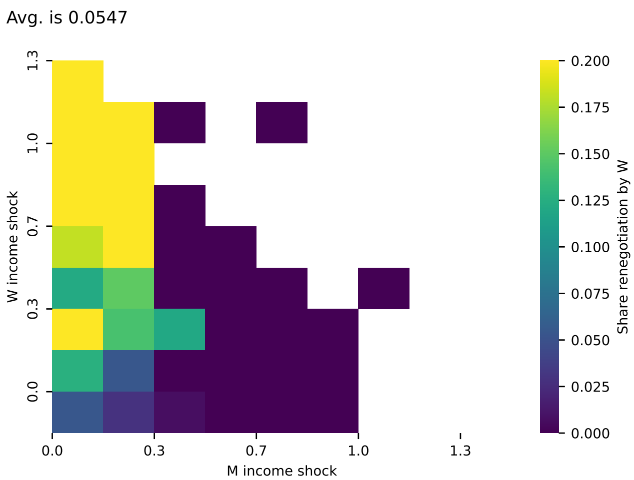
Figure 4: Share or divorces and renegotiations given relative earnings and match quality



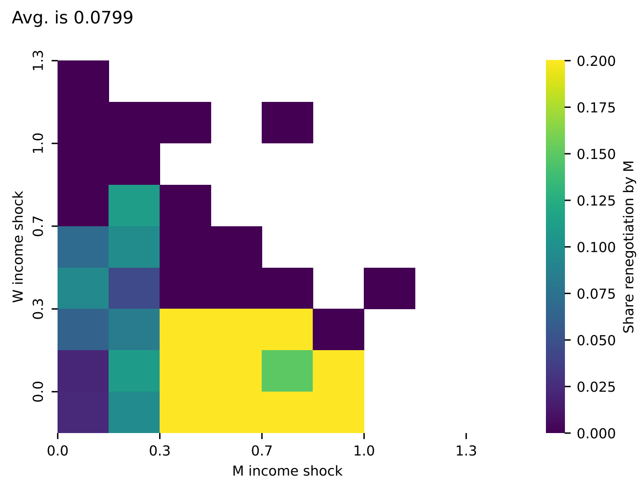
(a) Renegotiations and divorces



(b) Divorces



(c) Renegotiations triggered by the wife



(d) Renegotiation triggered by the husband

Figure 5: M and W income shocks, renegotiation and divorce

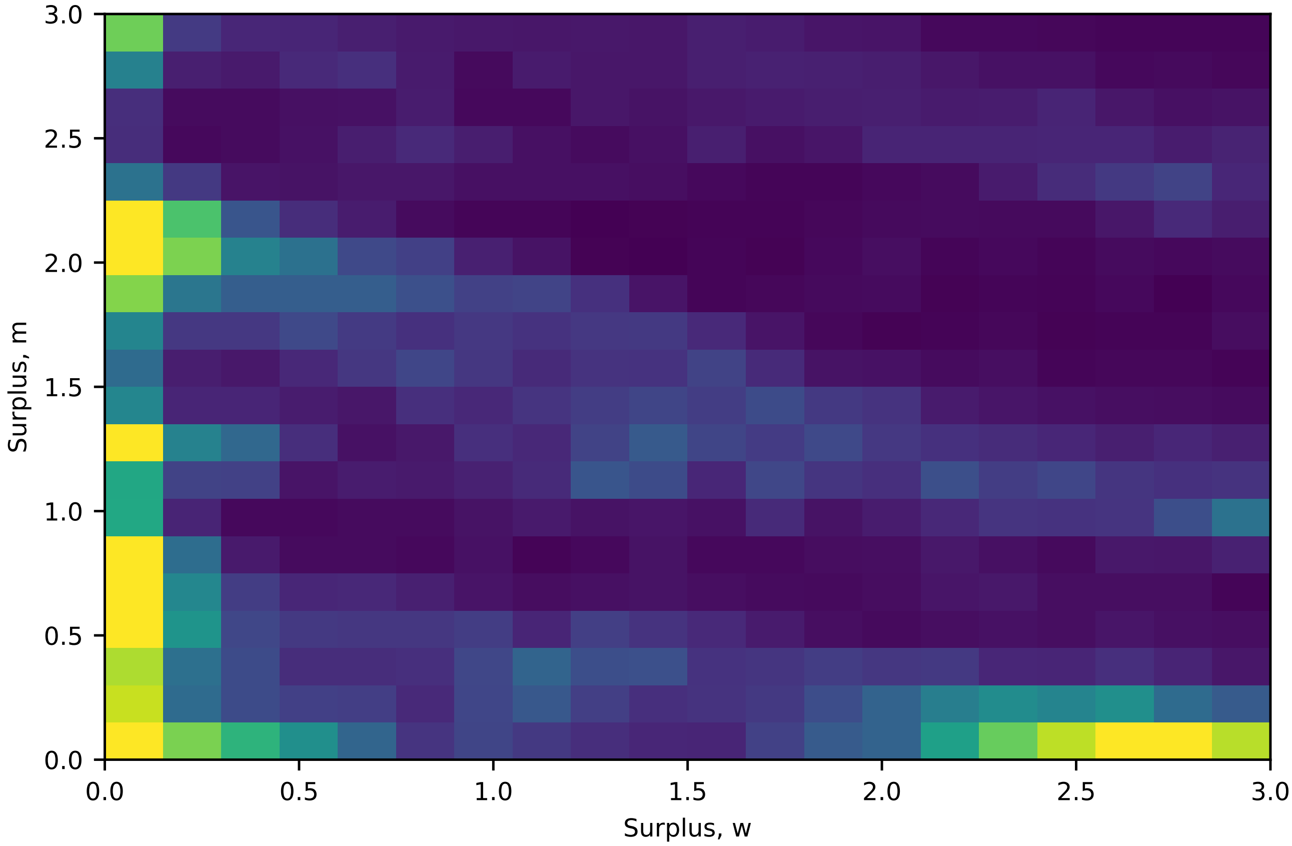


Figure 6: Marital surplus distribution (value of staying married - value of divorce)

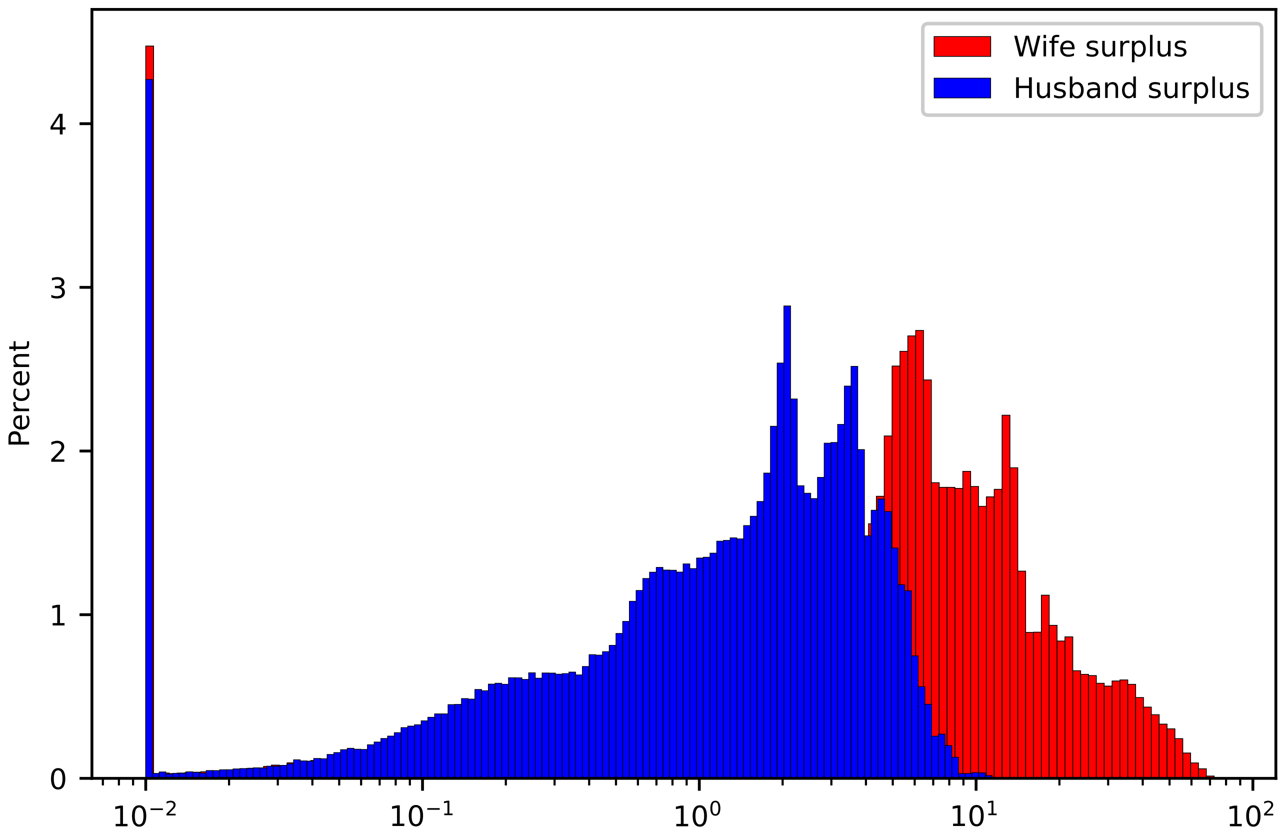


Figure 7: Marital surplus distribution

Figure 8: Marital surplus distribution at meeting

Something we have learned

- The match surplus at meeting is higher for women (Figure 8): this is an implication of the (close to) symmetric nash bargaining (SNB) and women earning less than men. SNB allocates a higher share of the surplus to the spouse having lower earnings.
- Women are more likely to hit the participation constraint than men (Figure 7). Since, again, women's marginal utility is higher than for men, the same shock implies a larger change in reservation utilities for women than for men.
- If we impose a non-symmetric nash bargaining, the gender who gets a higher weight will be less likely to hit participation constraints.
- If we close the gender wage gap, the patterns in renegotiation and surplus share distribution become gender symmetric
- To be checked with the policy experiment how labor supply is reacting. If, when outside option for women improve, labor supply goes down, our model cannot replicate it.
- Excess kurtosis of women consumption seems to be driven by renegotiations

Table 3: Pass-through of changes in income on consumption and consumption shares, using changes in...

	Total Exp (1)	Common Exp (2)	Husband Exp (3)	Wife Exp (4)	Wife share (5)
...total income	**0.250**	0.277			
...wife income	0.083	0.094	**0.001**	**0.099**	**0.098**
...husband income	0.155	0.162	**0.159**	**0.077**	**-0.083**

NOTES: Coefficient interpretation: 1% change in income leads to X% change in expenditure. Coefficients associated to changes in the wife income are computed using women working in two consecutive periods.

Table 4: Pass-through of changes in income on consumption and consumption shares, using **transitory** changes in...

	Total Exp (1)	Common Exp (2)	Husband Exp (3)	Wife Exp (4)	Wife share (5)
...total income	**0.086**	0.096			
...wife income	0.066	0.076	**-0.020**	**0.082**	**0.102**
...husband income	0.070	0.073	**0.071**	**0.041**	**-0.030**

NOTES: Coefficient interpretation: 1% change in income leads to X% change in expenditure. Coefficients associated to changes in the wife income are computed using women working in two consecutive periods.

4 Consumption insurance regressions

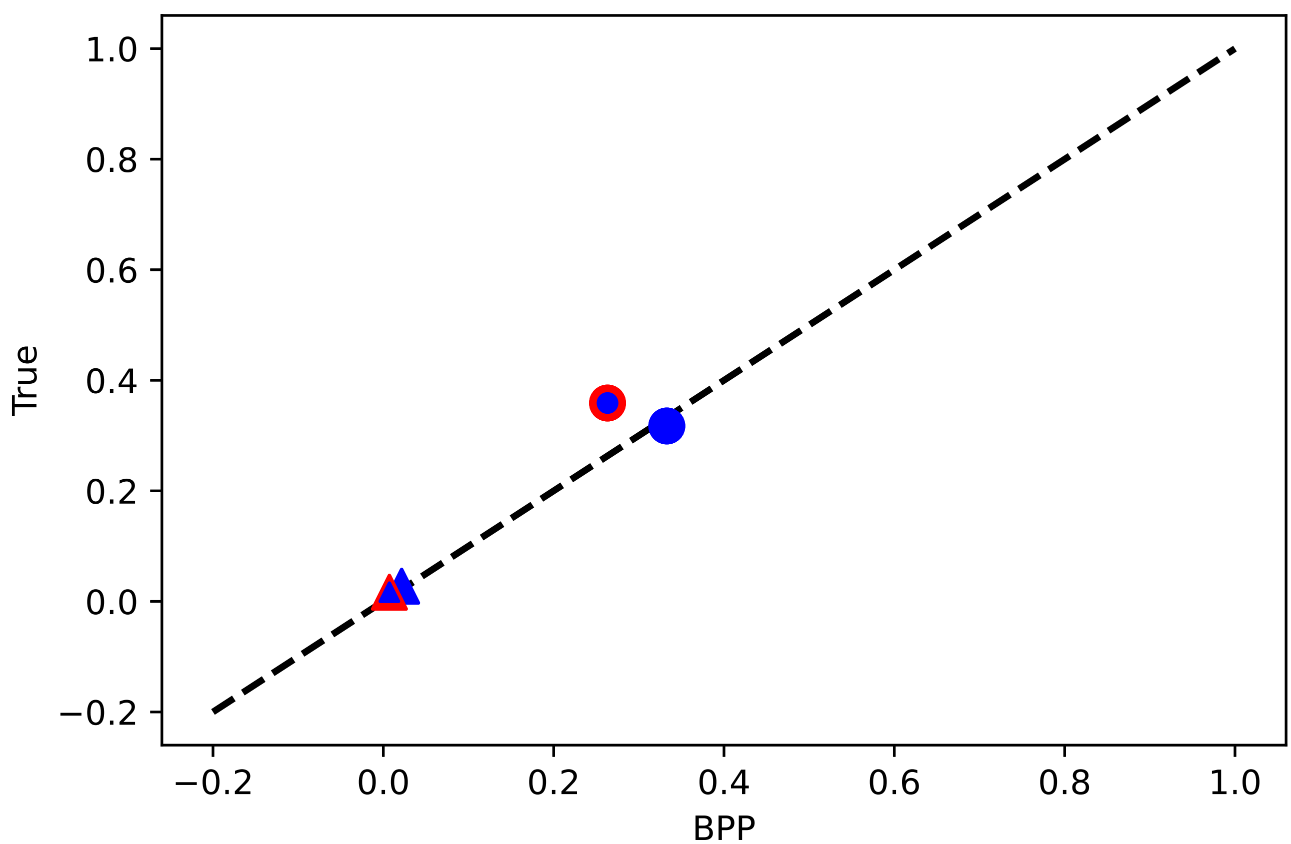


Figure 9: Pass-throughs: BPP vs. model-implied values. Circles (triangles) represent persistent (transitory) shocks. The outer (inner) color indicates the gender of the individual receiving the shock; the inner color corresponds to the individual whose consumption response is being measured.

Table 5: Pass-through of changes in income on consumption and consumption shares, using **persistent** changes in...

	Total Exp	Common Exp	Husband Exp	Wife Exp	Wife share
	(1)	(2)	(3)	(4)	(5)
...total income	**0.353**	0.375			
...wife income	0.455	0.484	**0.182**	**0.568**	**0.386**
...husband income	0.392	0.412	**0.424**	**0.156**	**-0.268**

Notes: Coefficient interpretation: 1% change in income leads to X% change in expenditure. Coefficients associated to changes in the wife income are computed using women working in two consecutive periods.

Table 6: MPC calculated as in BPP, using transitory changes in...

	Total Exp	Common Exp	Husband Exp	Wife Exp
	(1)	(2)	(3)	(4)
...husband income	0.056	0.058	0.072	0.014
...wife income	0.045	0.055	-0.014	0.035
...total income	0.168	0.191	0.081	0.082

Notes: the consumption insurance parameters displayed in the table are computed as

$$\frac{E\left(\Delta c_t \Delta y_{t+1}\right)}{E\left(\Delta y_t \Delta y_{t+1}\right)},$$

where y_t can the income of the husband, wife or the sum of the two (total). Variables c_t can be the total, common, husband or wife' expenditures. Coefficients associated to changes in the wife income are computed using women working in two consecutive periods.

Table 7: Consumption insurance to persistent income shocks, calculated as in BPP, using persistent changes in...

	Total Exp	Common Exp	Husband Exp	Wife Exp
	(1)	(2)	(3)	(4)
...husband income	0.479	0.505	0.476	0.223
...wife income	0.326	0.347	0.096	0.452
...total income	0.658	0.702	0.483	0.498

Notes: the consumption insurance parameters displayed in the table are computed as

$$\frac{E\left(\Delta c_t \left(\Delta y_{t-1} + \Delta y_t + \Delta y_t\right)\right)}{E\left(\Delta y_t \left(\Delta y_{t-1} + \Delta y_t + \Delta y_t\right)\right)},$$

where y_t can the income of the husband, wife or the sum of the two (total). Variables c_t can be the total, common, husband or wife' expenditures. Coefficients associated to changes in the wife income are computed using women working in two consecutive periods.

Table 8: Women's employment response (in percentage points) to different types of income shocks

Transitory shocks		Persistent shocks		Transitory+persistent shocks	
Wife	Husband	Wife	Husband	Wife	Husband
(1)	(2)	(3)	(4)	(5)	(6)
1.362	-0.182	1.058	-0.776	1.278	-0.323

Notes: the income shocks relate to *potential log income y*. In the case of women, a positive potential income shocks does not translate in more earnings if the women does not work. The numbers displayed in the table are OLS coefficients:

$$\frac{E(\Delta y_t \ \Delta WLP_t)}{E(\Delta y_t)},$$

where ΔWLP is the change in women's employment over two consecutive periods.

Table 9: Pass-through of changes in income on consumption and consumption shares, using changes in...

	Total Exp (1)	Common Exp (2)	Husband Exp (3)	Wife Exp (4)	Wife share (5)
...total income	**0.162**	0.141			
...wife income	0.173	0.157	**0.004**	**0.012**	**0.054**
...husband income	0.127	0.105	**0.018**	**0.004**	**-0.032**

NOTES: Coefficient interpretation: 1 yen change in income leads to X yen change in expenditure.

Table 10: Earnings to consumption pass-through, using...

	Husband Exp (1)	Wife Exp (2)	Common Exp (3)	Wife share (4)
...any husband shocks	**0.159**/0.229	**0.077**/0.167	**0.162**/0.161	**-0.083**/-0.010
...any wife shocks	**0.001**/0.069	**0.099**/0.076	**0.094**/0.028	**0.098**/0.066
...persistent husband shocks	**0.476**/0.610	**0.223**/0.243		
...persistent wife shocks	**0.096**/0.084	**0.452**/0.331		
...transitory husband shocks	**0.072**/0.191	**0.014**/0.212		
...transitory wife shocks	**-0.014**/0.128	**0.035**/-0.001		

NOTES: Simulations: **bold**; data: red.

Table 11: Individual Husband Consumption Insurance to Husband Persistent Shocks, decomposition

	Household consumption insurance						
	Wife labor supply						
	Passive	Active	Tax	Savings	Private Exp. share	Reneg.	Total Insurance
Baseline	27.7	15.1	4.2	11.4	8.0	-4.1	62.3
Alimony, low	25.4	16.2	3.7	14.7	7.0	-8.4	58.6
Alimony, high	24.5	17.7	3.6	11.7	7.2	-9.1	55.6

Table 12: Individual Husband Consumption Insurance to Husband Persistent Shocks, decomposition

	Household consumption insurance						
	Wife labor supply						
	Passive	Active	Tax	Savings	Private Exp. share	Reneg.	Total Insurance
A. Limited Commitment							
Baseline	27.7	15.1	4.2	11.4	8.0	-4.1	62.3
Low GWG	32.9	13.2	3.7	9.9	8.1	-5.0	62.7
No GWG	38.3	10.7	3.4	8.0	8.1	-6.0	62.5
B. Full Commitment							
Baseline	25.4	21.2	4.1	4.3	7.7	0.0	62.7
Low GWG	30.2	20.5	3.3	4.6	7.1	0.0	65.7
No GWG	36.3	16.0	2.9	6.0	7.0	0.0	68.2

NOTES: GWG = gender wage gap.